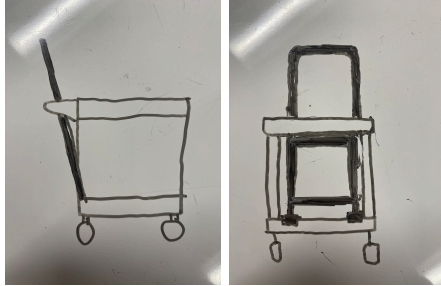


Element G

Final Prototype

The handle prototype that we have made is similar to the original design. The original design that we made is below.



We got the drawings for the original prototype and started by buying the connectors for the 1" PVC. We got four T connectors and two 90 degree connectors along with two end caps with open end clamps for $\frac{3}{4}$ " PVC. These allow for the handle to connect to the bottom of the cart.



With the connectors we started to measure and cut the PVC that we needed. We cut two 2" pieces to go between the bottom end clamps and two T junctions. Following that we measured and cut the horizontal pipe to the correct length so the bottom would slot on the cart. From that point we measured and cut the vertical pipes up to the bottom of the handle on the cart and connected those to two more T junctions with another horizontal pipe to increase the lateral stiffness.



Once we had the base of the handle built we worked with Mr. Kiser to get the length of the handle built to the right height. We then cut the pipes to the right length and connected them

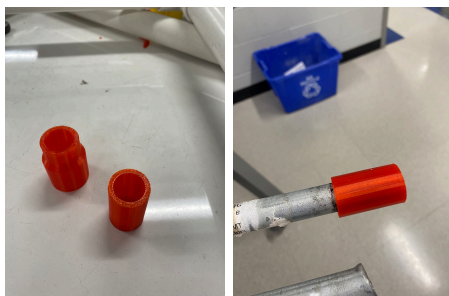
at the top with the 90 degree connectors and one more horizontal pipe.



From that point we started to test the handle and find ways that it needed improvement. Overall the handle worked well but the lateral stiffness was not as good as we desired. To combat this issue we decided to glue the 90 degree connectors and horizontal pipe to keep the handle from torquing the way it had been. This added resistance with the metal pipes that we are running through the handle should be enough to combat this issue.



Another issue that arose in the testing was the strength of the handle. After some testing one of the T junctions in the middle of the handle gained a small crack. This issue was larger as finding a solution was difficult without blowing through the budget and rebuilding the entire handle from scratch. After talking with several engineers we decided to use some small steel pipes that we got and 3D printing inserts so that they would fit tightly within the PVC. This solution removes the strain of pushing and pulling from the PVC and specifically the joint that had cracked. This allows for more strength and longevity of the handle with little added complexity and weight.



These changes created the final product we made for the handle design combining ease of use, strength, stiffness, and compliance. The final prototype works well as was originally intended.